The Detection Floor: What a Null Result Was Measured With

Allen Hsieh
FishAI project

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Abstract

A research codebase that publishes null results owes its readers a characterisation of the instrument those nulls were measured with, because an unresolved effect and an absent effect look identical in a table. The FishAI project published several nulls for the `us54` dialect of Literature (Canadian Fish) before anyone measured what its one instrument — K duplicate deal pairs of arm X against arm Y, reported as the mean paired set-difference $\pm 1.96\,\mathrm{SE}$ — could resolve. This paper reports that characterisation and applies it to the project's own back catalogue. The per-pair standard deviation is 3.15–3.44 sets for arms that differ at every decision (3.18–3.44 on an independent replication), giving an 80%-power minimum detectable effect running from 0.76 sets at $N = 150$ through 0.47 at $N = 400$, 0.33 at $N = 800$ and 0.14 at $N = 4300$. Validated against a planted edge of continuously variable known size, the normal-theory power model held in all twelve (effect, N) cells of both runs to within about one binomial standard error, and the byte-exact control — which had to be *changed* before it tested anything, an earlier form short-circuiting past the very wrapper whose inertness was the claim — prints 0.0000 ± 0.0000. Two parts did not replicate and are reported unsoftened: the empirical floor at $N = 400$ moved from 84.7% detection to 75.0% on fresh banks (pooled 81.7%), so the floor is only bounded to 0.36–0.50; and nominal-95% interval coverage at $N = 400$ went 6.1% (11/180) to 10.8% (13/120), pooling to 8.0% $[5.2, 11.7]$, which excludes the nominal 5% — while $N = 800$ is nominal in both runs, a pattern that does not scale as a real coverage defect should, so the cause is unknown. We name the trap the first run fell into: the minimum detectable effect depends on *that cell's own* standard deviation, which is not constant (3.645 for one cell; 4.57–4.95 implied by a foreign-engine harness), and a cell whose two arms mostly agree can resolve $+0.2475 \pm 0.1024$ at $n = 800$ while appearing to sit under a generic 0.33 floor. Applying the floor reclassifies two of this project's published nulls as non-measurements. The recommendation, grounded only in this codebase's experience, is to publish the floor beside the null.

1 Introduction

There are two reasons a measured effect can be reported as zero. The effect can be absent, or the instrument can be unable to see it. In a results table these look the same: a small number, an interval containing zero, and a row that reads as settled. The distinction is not pedantic. The first licenses abandoning a mechanism; the second licenses nothing at all except a bigger experiment. Altman and Bland put the point in five words that have not been improved on — absence of evidence is not evidence of absence [10].

This paper is about a codebase that made the mistake and then measured its way out of it. The FishAI project builds an engine, a bot laboratory and a results site for six-player Literature (Canadian Fish) in its 54-card `us54` dialect [8]. Its research documents are unusual in one respect that makes the failure especially visible: they publish negative results as first-class findings, in the body text, with the same typography as the positive ones [1, 7, 6]. Four of the project's six papers at the time of writing carry a negative headline.

Every one of those headlines — positive and negative alike — comes out of a single instrument: K duplicate deal pairs of arm X against arm Y under `us54`, reported as the mean paired set-difference $\pm 1.96\,\mathrm{SE}$. Until the work reported here, nobody had characterised what that instrument can and cannot see. Every published null was therefore ambiguous between "no effect" and "no resolution", and the project had no principled way to tell its readers which it had found. Worse, the ambiguity is asymmetric in exactly the direction that flatters a project that prides itself on negative results: an underpowered design manufactures publishable-looking nulls for free.

The remedy is a scratch harness, `scripts/probe-floor.mjs` [2], and an independent agent's re-run of the whole thing on fresh seed banks. Together they answer four questions — what is the noise level, what is the smallest planted effect the design actually detects, how often does a true null print an interval excluding zero, and which committed numbers sit below their own floor. The results are recorded in [1] §8a, which this paper reports.

Three of the four parts replicated. Two did not, and both cut against the first run's conclusions. This paper reports the failures at the same size as the successes, because a floor paper that softened its own non-replications would be a self-refuting document.

The paper is organised as follows. Section 2 states the instrument precisely. Section 3 gives the noise level and the resulting power curve. Section 4 describes the planted-edge validation, and Section 5 — the methodological heart of the paper — describes why the byte-exact control had to be changed before it was worth anything. Section 6 reports the two parts that did not replicate. Section 7 names the substitution trap that the first run fell into. Section 8 applies the floor to the project's committed results and reclassifies two of its published nulls. Section 9 states what none of this establishes, and Section 10 makes the one recommendation the evidence supports.

2 The instrument

2.1 What a datum is

Literature is played by six players in two teams of three, seats $0, 2, 4$ against $1, 3, 5$ [8]. Under `us54` the 54-card deck is partitioned into nine books of six cards; the game ends the moment a team is awarded five books. A game therefore terminates in a clinch, and the outcome variable used throughout the project's concession research is not the win indicator but the *net sets*: the acting team's score minus the opposing team's.

The instrument is duplicate pairing, borrowed from duplicate bridge and used so that the deal is never a confound [5]. Every seed is played twice, once with arm X on team 0 and once with X on team 1, both arms otherwise identical. The pair's datum is

$$d_i \;=\; \big(X - Y \mid X \text{ on team } 0\big) \;+\; \big(X - Y \mid X \text{ on team } 1\big), \tag{1}$$

on the same seed, and the reported statistic over K pairs is $\bar{d} \pm 1.96 \cdot \mathrm{SD}(d)/\sqrt{K}$ [2]. The constant 1.96 is a house constant, fixed so that every interval in the repository is comparable to every other.

Two properties of this design bear on everything below. First, a *mirror match* — the same arm against itself — returns exactly zero for every pair by construction, since the two orientations cancel term by term. That is a useful correctness check and a useless noise estimate: it tells you nothing about the variance of a comparison between arms that actually differ. Second, the pairing removes the deal but not the play. Two arms that differ at any decision produce different game trees on the same deal, and the residual variance of d_i is what the floor is made of.

2.2 The four questions

`probe-floor.mjs` [2] answers four questions, in four stages that can be run independently:

1. `sd` — the per-pair standard deviation under the conditions the published tables actually used, and the analytic minimum detectable effect (MDE) curve computed off it. The default is 1,200 pairs per condition.

2. `calib` and `detect` — the *empirical* floor. An edge of continuously variable true size is planted, measured at large N (2,000 pairs per grid point), and then the fraction of independent banks whose nominal 95% interval excludes zero is counted at $N = 400$ and $N = 800$. The default is 36 banks of 800 pairs per planted size.

3. `fpr` — the false-positive rate, on true nulls that nevertheless move the game.

4. `published` — the floor applied to the numbers already committed.

Every seed bank the probe uses is a fresh prefix under `floor-`, disjoint from all twelve banks in use elsewhere in the repository. The single exception is deliberate and is described in Section 5.

2.3 The power arithmetic

The MDE is the standard two-sided normal-theory quantity for a paired mean at 80% power:

$$\mathrm{MDE}(N) \;=\; \frac{\left(z_{0.975} + z_{0.80}\right)\mathrm{SD}}{\sqrt{N}} \;=\; \frac{2.8016 \cdot \mathrm{SD}}{\sqrt{N}}, \tag{2}$$

with $z_{0.975} = 1.96$ the house constant and $z_{0.80} = 0.8416$ the one-sided power point [2, 11]. Nothing about equation (2) is novel; the contribution here is that the SD in it was, in this repository, never measured before it was needed, and that when it was measured it turned out not to be a constant (Section 7).

3 The noise level, and the curve that follows

3.1 Per-pair standard deviation

Table 1 gives the per-pair SD for each condition the probe measures, as recorded in the harness's own `MEASURED_SD` constant — the values its `sd` stage returned on 2026-08-31 at 1,200 pairs per condition, checked in so the `published` stage can be re-run without paying for the sweep again [2].

The headline is the range over the eight full-divergence conditions: 3.15 **to** 3.44 **sets per pair**. The independent replication on fresh banks returned 3.18–3.44 [1]. This is the most stable

Table 1: Per-pair SD of the duplicate-pair difference, by condition, 1,200 pairs each [2]. The first eight rows are arms that differ at every decision; the last two are arms that agree almost everywhere, and are included here because Section 7 turns on the difference.

condition	per-pair SD (sets)
defuse, balanced ([1] §3.1)	3.321
defuse, ghost (§3.1, narrowest)	3.237
defuse, archivist (§3.1, widest)	3.428
conceal vs. defusing opponent (§5a.3)	3.428
conceal vs. non-defusing opponent (§5a.3)	3.363
lopsided: balanced vs. turtle	3.438
gauntlet difference-of-differences vs. balanced (§3.3)	3.364
information-free null, $\lambda = 0.25$ (§2.1)	3.150
planted edge, $\varepsilon = 0.02$	2.549
symmetric null, $\varepsilon = 0.02$	2.959

result in the whole exercise, and it is worth pausing on why it should be stable: it is a property of the game's outcome distribution under duplicate pairing, not of any particular mechanism, and it varies remarkably little across conditions as different as a one-knob ablation and a lopsided style mismatch.

3.2 The curve

Substituting into equation (2) gives Table 2, the operative design tool of this paper.

Table 2: 80%-power minimum detectable effect, in sets per duplicate pair, at the measured per-pair SD [1].

N (pairs)	150	300	400	600	800	2000	4300
MDE (sets/pair)	0.76	0.54	0.47	0.38	0.33	0.21	0.14

Read it as a constraint rather than a licence. At $N = 400$ — the size at which several of this project's cells were run — an effect of a third of a set is more likely than not to be missed. At $N = 150$, the size of the cross-engine cells in [3] §2, the design cannot resolve anything under three quarters of a set.

4 Validating the model: an edge of known size

A power curve computed from a measured SD is a prediction, not a measurement. It assumes normality of $\bar{d}$, correct SD estimation, and independence across pairs. The probe therefore checks the prediction against an edge whose true size is known because it was put there.

4.1 The planted edge

The handicapped arm plays the shipped policy except that, on an ordinary ask, it replaces the chosen ask with a uniformly chosen legal alternative with probability ε [2]. Two properties make this a usable ruler:

- at $\varepsilon = 0$ no random draw is ever taken — `u01` returns a value in $[0, 1)$ and every such value fails the $\geq \varepsilon$ test — so the arm is byte-identical to the control and the paired difference must print exactly 0.0000 ± 0.0000;
- the true edge grows monotonically with ε, because replacing an argmax by a uniform draw over the legal ask list can only lose value in expectation.

The draw is a deterministic hash of (stream, seed, step, seat), so a bank replays identically and the two orientations of a pair remain the same deal — which is what keeps equation (1) meaningful under the handicap. The calibration grid is $\varepsilon \in \{0.005, 0.01, 0.02, 0.03, 0.04, 0.06, 0.08\}$, measured at 2,000 pairs per point so that the true edge at each ε is itself known far more precisely than the $N = 400$ and $N = 800$ banks that are then asked to detect it.

4.2 Two nulls that move the game

A mirror match is exactly zero by construction (Section 2), so it cannot serve as the null against which false positives are counted. The probe uses two true nulls that *do* disturb play [2]:

- `symnull` — both arms carry the *same* handicap magnitude ε, on two different hash streams. They differ at roughly 2ε of all ask decisions, yet the expected edge is exactly zero by the exchangeability of the two streams. This is the cleanest null available: the game moves, and the truth is nonetheless known.
- `abnull` — the information-free control of the project's earlier probe `probe-ab.mjs`, reused as specified: the danger penalty's magnitude with the threat estimate D replaced by a hash of (seat, log length) over the same 0–3 prey range. The prototype ranker is copied faithfully, so that appetite $\lambda = 0$ reproduces `decide` at `defuse: 0` byte-for-byte. That is a weaker statement than reproducing the shipped decision function, and the difference matters: the shipped roster carries `defuse: 1`, so what this control certifies is fidelity to the *unequipped* baseline, not to the policy the project ships [2].

In the `fpr` stage `symnull` is run at two magnitudes, $\varepsilon = 0.02$ and $\varepsilon = 0.08$, and `abnull` at the single appetite $\lambda = 0.25$ that `probe-ab.mjs` used [2]. All three arms are run at banks of 800 pairs each split into its two disjoint halves of 400, so that each bank yields one estimate at $N = 800$ and two at $N = 400$.

4.3 The power model holds

This is the solid finding of the exercise. Across twelve (planted-effect, N) cells — six planted sizes at two sample sizes — the observed detection rate sat within about one binomial standard error of the normal-theory prediction in *both* runs [1]. Equation (2) can therefore be trusted as a design tool: an experimenter in this repository who wants 80% power against a half-set effect can read off the required N and expect to get what the arithmetic promises.

What the model does *not* deliver is a tight empirical estimate of where the floor sits, which is Section 6's subject and a different question. The model being calibrated in aggregate is compatible with the single grid point nearest the floor being estimated badly, and that is what happened.

5 The control that had to be changed before it meant anything

The project's probe discipline requires a byte-exact control: an arm which, with the mechanism switched off, reproduces the shipped policy exactly, so that the harness prints 0.0000 ± 0.0000 and any nonzero reading elsewhere is attributable to the mechanism rather than to the scaffolding. Every published concession result in [1] carries one.

The floor probe's control is the planted-edge arm at $\varepsilon = 0$. An earlier version of that arm opened with a short-circuit:

```
if (!(eps > 0)) return plainPolicy(style)
```

which is the natural thing to write and is worthless as a control. With the short-circuit in place the control returns before the wrapper does anything, so the comparison it certifies is `plainPolicy(style)` against the shipped policy — a statement true by inspection, since `plainPolicy` is a one-line delegation to `decide`. The wrapper whose inertness at $\varepsilon = 0$ is the actual claim is never entered. The draw is never taken, the legality lookup never runs, the substitution branch is never reached, and a bug in any of the three would sail through a control printing a perfect zero [2].

The fix is to delete the short-circuit and let the arm fall through. The arm then takes the draw, computes the legal ask list, evaluates the substitution branch's guard, declines because $u01 \geq 0$ always holds, and *still* prints 0.0000 ± 0.0000 over 200 pairs. That is the statement the control discipline was asking for. The harness's own comment states the point plainly, and the file preserves the superseded version in prose so the reasoning survives the diff.

Three observations generalise beyond this file.

A control must exercise the code whose inertness it certifies. The failure mode is not that the short-circuited control was wrong — its output was correct — but that it was *vacuous*: it verified a proposition nobody doubted while appearing to verify one that mattered. A green control is not informative unless one can say which lines it ran.

The fast path and the null path are the same path here. Optimising a null case by returning early is normally harmless and often good practice. It is specifically harmful when the null case is the control, because the optimisation removes exactly the coverage the control exists to provide. Any codebase whose experimental controls are "the mechanism at appetite zero" has this hazard structurally.

The fidelity check is a separate obligation. A control shows the scaffolding is inert; it does not show the scaffolding measures the same thing the published harness measured. The probe therefore adds a fourth control job that deliberately breaks its own fresh-bank rule and replays `holdoutA`, the bank on which [1] §3.1 published $+1.4250 \pm 0.3179$ over 400 pairs for the shipped defusal mechanism. Its purpose is to reproduce a published number rather than to measure a new one, and that is the only reason a probe should ever reuse a bank [2].

6 The two parts that did not replicate

An independent agent re-ran the entire probe on fresh seed banks. Two of the four parts came back different, and both differences remove a claim the first run had made.

6.1 The empirical floor at $N = 400$

The first run found that a planted true effect of 0.393 sets per pair was detected — its nominal 95% interval excluding zero — in 84.7% of independent banks at $N = 400$. That is comfortably above the 80% target, and the first run drew the conclusion that the empirical floor *matched the analytic MDE to 1.5%*: a satisfying result, and the kind of agreement between prediction and measurement that invites trust in both.

The replication detected the same planted effect in 75.0% of banks. Pooled over both runs, the detection rate at that grid point is 81.7%. Table 3 sets the two runs side by side.

Table 3: The two non-replications [1]. Every figure is as printed in the source; the interval construction used for the coverage rows is not recorded alongside them.

quantity	first run	replication	pooled
detection of a 0.393 effect at $N = 400$	84.7%	75.0%	81.7%
empirical floor at $N = 800$	0.256	0.275	replicates
coverage: intervals excluding zero, $N = 400$	6.1% (11/180)	10.8% (13/120)	**8.0% (24/300)**
95% CI on that rate	—	$[5.9, 17.8]$	$[5.2, 11.7]$
coverage at $N = 800$	nominal	nominal	4.0%

The consequence is that **the $N = 400$ floor is bounded only to the interval 0.36–0.50 sets per pair, and this design cannot pin it tighter**. The first run's 1.5% agreement was an artifact of where the sweep grid happened to land: with a grid of planted sizes and a detection rate estimated from a few dozen banks per point, the identity of the smallest point clearing 80% is a coarse and unstable statistic, and reading a two-significant-figure agreement off it was over-reading. The analytic MDE at $N = 400$, 0.47, lies inside the bracket, which is reassuring and is not the same as being confirmed.

The $N = 800$ floor *does* replicate: 0.256 against 0.275. The design is better behaved where it has more data, which is what one would expect and is a reason to prefer the $N = 800$ column of Table 2 when a choice is available.

6.2 Interval coverage at $N = 400$

This is the more uncomfortable of the two, because it removes the first run's strongest claim.

On true nulls that still move the game (Section 4.2), the first run counted 11 of 180 nominal-95% intervals excluding zero at $N = 400$: 6.1%, which is nominal. On that basis the first run concluded that *the nominal 95% intervals are correctly sized; no interval in this repository needs widening*. That is a strong and useful claim, and it would have retired an entire class of worry about the project's published error bars.

The replication counted 13 of 120: 10.8%, with a 95% interval of $[5.9, 17.8]$ that excludes 5%. Pooled over both runs the rate is **8.0**% (24/300), with interval $[\mathbf{5.2}, \mathbf{11.7}]$ — which also excludes the nominal 5%. **The first run's strongest claim fails on independent banks.**

At $N = 800$ both runs are nominal, pooling to 4.0%.

We report the cause as **unknown**, and say why rather than leaving it at a shrug. A genuine under-coverage defect — a heavy-tailed pair distribution, a variance estimate biased low, a residual dependence between the two orientations of a pair — ought to scale with N in a recognisable way: worse at smaller N, tending to nominal as the central limit theorem takes hold. What is observed is nominal at $N = 800$ and anomalous at $N = 400$, from banks that are literally the halves of the $N = 800$ banks. That is a very specific pattern and no mechanism proposed so far predicts it. The two candidate readings are that $N = 400$ is genuinely below the size at which the normal approximation is safe for this distribution and $N = 800$ is above it — a sharp transition over one doubling, which is a lot to ask — or that the excess is a fluctuation and 24/300 with an interval whose lower end sits at 5.2% is simply a near-miss dressed as a finding.

We decline to choose. What can be said without choosing is the practical instruction: **a nominal 95% interval computed at $N = 400$ in this repository should not be treated as a 95% interval**, and the honest reading of a marginal $N = 400$ result is that its error bar may be optimistic by an unquantified amount. Whether this should be assumed benign is exactly the question [1] §9 refuses to close.

7 The substitution trap

The first run of this probe fell into a trap worth naming, because it is the most transferable mistake in the paper and because it is invisible if one reads Table 2 as a threshold rather than as a formula.

7.1 The SD is not a constant

Equation (2) contains a standard deviation. Table 2 was computed with one particular value of it, from one particular family of conditions. The MDE of any given cell depends on *that cell's own* SD, and the SD is not constant across this repository:

- The eight full-divergence conditions of Table 1 span 3.15–3.44, which is narrow enough to invite the substitution.
- The Turtle cell of [1] §3.1 measures **3.645** — outside that band, in the direction that makes its floor higher than the generic figure suggests.
- The cross-engine harness of [3] §3, which runs FishAI's policy inside a third-party engine against a foreign bot family, publishes half-widths of 0.517, 0.555, 0.533 and 0.560 at 300 duplicate pairs per opponent. Those imply a per-pair SD of **4.57** to **4.95**: **38–49% higher** than the `us54` figure [1].

Substituting one SD across all conditions understates the floor exactly where the cells are smallest, which is exactly where the understatement does the most damage. The cross-engine cells are the clearest case: they are both the noisiest and the shortest-run measurements in the project.

7.2 The counter-example that keeps the reader honest

The trap has a mirror image, and it matters more, because it is the error a reader makes with Table 2 in hand and good intentions.

A head-to-head between two adaptive generations, *v1.0 against v0.5* — two members of the same generational ladder [9] — is recorded as $+0.2475 \pm 0.1024$ at $n = 800$. Against the generic 0.33 floor at $N = 800$ that reads as "below the floor, not resolved" — and the reading is wrong. Its own interval excludes zero comfortably, by a margin of more than two half-widths. The reason is structural: the generic floor was computed from arms that differ at *every* decision, and two adaptive arms that mostly agree produce a far smaller per-pair SD than a one-knob ablation does, so their cell's own floor is far lower than the generic one.

Table 1's last two rows are the committed evidence for that mechanism inside this harness. The planted-edge arm at $\varepsilon = 0.02$ — which differs from its control on about 2% of ask decisions — has a per-pair SD of 2.549, and the symmetric null at the same magnitude 2.959, against 3.15–3.44 for arms that differ everywhere. Agreement between arms buys precision.

The rule that follows is short and is stated in [1] in bold for a reason: **never read the floor table as a threshold to apply by eye**. The floor is a function with an argument, and the argument is the cell's own noise.

7.3 The counter-example is itself not traceable

Honesty about a floor paper requires one more sentence about that counter-example, and it is not a comfortable one. **The figure $+0.2475 \pm 0.1024$ cannot be found in any committed document.** It appears exactly twice in the repository: in [1] §8a, where it does the work described above, and in the probe's own list of published claims, where its provenance field reads `'task brief'` and its description carries the parenthesis *"NOT FOUND in any committed doc"* [2].

We report this rather than quietly substituting a traceable example, for three reasons. It is the project's own rule that every number must be traceable to a committed source, and this one is not. The probe's author noticed and labelled it, which is the behaviour the rule is meant to produce, and suppressing the label would waste that. And the pedagogical point the number carries — that two mostly-agreeing arms resolve effects a generic floor would reject — does not depend on it: the mechanism is demonstrated inside the harness by Table 1's last two rows, on banks the probe generated and can regenerate. The counter-example should be re-derived from a cell that exists in a committed artifact, and until it is, it should be read as an illustration rather than as a measurement.

8 Applying the floor to the back catalogue

The point of characterising an instrument is to go back and re-read what it produced. Table 4 is [1] §8a.1: every headline result of the project's concession research placed against its own floor.

8.1 Two nulls become non-measurements

The two reclassifications are the substantive output of this exercise, and both are corrections to claims this project published.

[1] §6's "inert" contained-pass aim. The project replaced the aim and cost model of its contained-book turn-pass — the one place in the engine where "concede to the least dangerous seat" is exactly right, because a contained-book pass is a guaranteed miss by construction and so cannot suffer the danger/opportunity confound that sinks the same rule on ordinary asks — with the measured threat estimate, and got -0.045 ± 0.130 over 400 duplicate pairs. The reading is now **a statement about this instrument's resolution and not about the game**: at the $N = 400$ that cell was run at, the floor is 0.47 sets per pair, so an effect of 0.045 is an order of magnitude under the only floor that applies to it — and it stays a factor of three under 0.14, the smallest floor anywhere in Table 2.

The reclassification is not a call to un-revert, and it is worth being exact about what it does and does not touch, because the reason §6 gives for the revert was never the interval. [1] §6 states

Table 4: Committed results against their own detection floor [1]. "Its floor" is the 80%-power MDE at that cell's N and, where it could be measured, that cell's own SD.

result	reported	N	its floor	verdict
§5a.3 concealment vs. non-defusing opponents	-0.1483 ± 0.2571	600	$\sim$0.38	**below floor — not resolved**
§6 the contained-pass aim, "inert"	-0.045 ± 0.130	—	—	**below floor — a non-measurement, not a zero**
[3] §2, all four headline cells	42.7–48.7%, no intervals published	150	1.05–1.13	**all four below floor**
§3.1 defusal, shipped	$+1.43$ / $+1.58 \pm 0.32$	400	$\sim$0.47	resolved, 3× the floor
§8.1 the six triangle edges	$\pm$0.10 half-widths	4000	$\sim$0.15	five resolved; P vs. C (+0.1457, +0.1742) sits at the floor — **nominal only**
[4] §5a, v1.5 vs. v1.0	-0.1255 ± 0.0590	see §8.2	see note	resolved — that cell's arms differ far less, so its own SD is much smaller

that the change was **reverted rather than shipped off by default**, on that document's own §6.2 argument: a knob that is listed, swept and reported while being inert contaminates the measured vector, and one such knob is already one too many [6]. That reason stands untouched here. Two further things should be said in the reverted change's defence. First, the diagnostic that accompanied the null gives an independent mechanism for it: over 25,654 ask decisions, more than one opponent still held cards 99.1% of the time, the contained pass fired at all on only 1.06%, and it fired *and* the two aims disagreed on 0.11% [1]. A mechanism that gets to matter on one decision in a thousand cannot move a set-difference, and that argument does not depend on the interval. Second, the reverted aim was the soundest available form of the fix, built at the one place in the engine where the danger/opportunity confound cannot arise. What the reclassification removes is only the supporting reading — the word "inert" — and not the ground the revert was actually made on.

[1] §5a.3's non-defuser cell. Concealment measured -0.1483 ± 0.2571 at $N = 600$ against opponents that do not defuse, and the document originally read that as concealment being worth nothing against them. At that cell's own SD of 3.363 the floor is about 0.38 sets: the measurement is not capable of distinguishing a real loss of about a third of a set from a small gain. The honest statement, now in the source document, is that the cell is **not resolved**. What survives is the comparative result measured at 4,000 pairs per cell, where concealment-only is the weakest of the four concession arms outright — a claim whose floor is about 0.15 and which clears it.

And a third that is not this project's own. [3] §2's per-opponent ranking of FishAI against a foreign bot family — four headline cells at 42.7% to 48.7% win rate, at 150 games per cell, published without intervals — sits under a floor of 1.05–1.13 sets per pair on that harness's own noise level. The ordering of those four opponents is not resolved. The *aggregate* claim that harness was built to make, that defusal replicates in a foreign engine at +1.55 to +1.75 sets per pair over 300 pairs per opponent with four intervals excluding zero, is untouched by this.

8.2 Two discrepancies found while reproducing the table

Reproducing Table 4 against its underlying sources surfaced two bookkeeping discrepancies. Neither changes a verdict; both are reported because a floor paper that quietly normalised its own inputs would be exactly the kind of document it is arguing against.

The bounded-memory row's N. [1] §8a.1 records the v1.5-against-v1.0 cell at $N = 2700$. [4] §5a, the document that published it, states 2,000 duplicate pairs per rung on the held-out bank, and the probe's own list of published claims carries $n = 2000$ for the same row [2]. Two committed sources say 2,000 and one says 2,700; the balance of evidence is that the table's N is a transcription error. The row's verdict does not turn on it, because that row is resolved for a reason about its SD rather than its N (Section 7.2) — which is the only reason this is a footnote rather than a correction.

The untraceable counter-example, Section 7.3.

8.3 A verdict rule worth stating separately

The probe's `published` stage does not treat "the published interval excluded zero" as sufficient for "resolved". It requires both that the interval excluded zero *and* that the reported effect is at or above the 80%-power floor, and it prints a distinct verdict — `NOMINAL ONLY -- underpowered` — for a cell that clears its own interval but not its floor [2].

The reasoning is standard and under-applied. An effect that clears its interval but sits below the design's floor was found by a design that would have missed it more often than not; conditional on being found at all, such an estimate is biased upward in magnitude, and if the true effect is small its sign is not reliable either [12]. Reporting it as "significant" without reporting that its design had, say, 40% power against it is how a literature accumulates effect sizes that shrink on replication [13, 14]; the same argument is why the reform literature asks for estimates and intervals to be read as continuous evidence rather than sorted into significant and not [15]. Naming the underpowered-but-significant case in the harness's own output, rather than in a caveat somewhere else, is the cheapest available guard against it.

The rule has a cost, and this paper pays it on its own table rather than exempting itself. Applied to §8.1's six triangle edges, five clear the ~ 0.15 floor at $N = 4000$ by a wide margin, but the weakest — P against C, plain against conceal-only — is $+0.1457 \pm 0.1030$ on the first bank set and $+0.1742 \pm 0.1030$ on the replication [1]. Both intervals exclude zero; neither effect is three times its floor, and the first sits fractionally *under* it. By this paper's own rule that edge is **nominal only, not resolved**. Nothing about the triangle's refutation depends on it — that argument is carried by wrong-signed legs many times their floor — but the ordering claim that conceal-only is the worst of the four arms rests on this edge, and it rests on it at the floor.

9 What this does not establish

The floor is a property of this instrument on this game. 3.15–3.44 sets per pair is the residual noise of duplicate-paired net sets under `us54` at this roster's level of play. It is not a property of Literature, of duplicate pairing in general, or of any other outcome variable. The cross-engine harness's 4.57–4.95 is the nearest available demonstration that the number does not transfer even to a different implementation of the same game.

The floor applies to the paired set-difference and to nothing else. Several of this project's nulls are reported in other units. The contained-book paper's headline mirror table, for instance, reports mean score on a 0.5-centred scale with standard errors of 0.0000–0.0065 [7]; that is a different statistic on a different scale, and Table 2 says nothing about it. A floor characterisation is per-instrument, and this project has more than one instrument.

Nothing here rescues a null that is below its floor. Reclassifying §6 and §5a.3 as non-measurements does not make either mechanism valuable. It removes a claim; it does not install the opposite claim. Both cells would need re-running at an N chosen against the effect size actually worth detecting, and neither has been.

The empirical floor at $N = 400$ is bracketed, not measured. Section 6.1's 0.36–0.50 is the honest width of what two runs of this design can say. Pinning it tighter needs either many more banks per grid point or a finer grid, and the exercise would have to be re-costed against what a tighter number would actually buy.

The coverage anomaly is unresolved and may be nothing. Section 6.2 states both readings and chooses neither. A reader who wants a single number should take $N = 800$'s 4.0%, use $N \geq 800$ for anything marginal, and treat $N = 400$ intervals as provisional.

The probes are not the lab. `probe-floor.mjs` and its siblings are scratch harnesses: not typechecked, emitting no committed artifact, and their numbers are not digest-bound [1]. They exist so that every table can be regenerated, and that is a weaker guarantee than the one the lab's committed results carry. The one mitigation is the fidelity check of Section 5, which shows this harness reproduces a published number on a published bank.

Decision-level identity is a stronger tool where it applies. Where the question is whether two policies differ *at all*, an exact comparison of their actions on identical observations and seeds returns a certificate rather than an interval, and no floor arithmetic is needed [6]. The floor characterised here is the right instrument for "how much is this worth" and the wrong one for "does this fire". The project has both, and part of reading a null correctly is noticing which question was asked.

10 The recommendation

One recommendation follows from this experience, and it is deliberately modest, because it is grounded in one codebase, one game, and one instrument.

Publish the floor beside the null. A reported null should carry, in the same table row, the smallest effect the design that produced it would have detected 80% of the time — computed with that cell's own noise level, not a repository-wide constant. Nothing else in this paper is a recommendation to anyone. This one is, for four reasons the work made concrete:

1. **It is the cheapest possible remedy for the ambiguity.** The floor is one number, computed from a standard deviation the harness has already estimated in the course of printing its interval. It requires no additional games. Two of this project's nulls were misread for want of a number that was, in effect, already in memory.

2. **It changes what the null licenses.** "Measured at -0.045 ± 0.130" invites the reader to abandon the mechanism. "Measured at -0.045 ± 0.130, against a floor of 0.47" invites the reader to note that nothing was learned and to ask what N would be needed. Those are different actions, and only the second is supported.

3. **It has to be per-cell, and stating it per-cell is what stops the substitution trap.** A repository-wide floor printed once in a methods section becomes a threshold applied by

eye, which is precisely the error of Section 7. Printed in the row, beside its own interval, the floor carries its argument with it.

4. **It composes with the discipline of putting the correction where the claim is.** This project's practice is to correct claims adjacent to the claims themselves rather than in a footnote or a later document [6, 1]. A floor in the row is that practice applied pre-emptively: the caveat cannot become detached from the number it qualifies, because it is part of the number's presentation.

The negative form of the recommendation is worth stating too, because this paper is an instance of it. **Do not publish a floor without replicating it.** The first run of this exercise produced a clean, quotable, mutually confirming set of four results, and one independent re-run on fresh banks removed two of them — including the strongest. Had the floor been published after one run, this project would have added a false reassurance about its own error bars to the pile of claims it was trying to clean up. A characterisation of an instrument is itself a measurement, and it inherits every obligation the measurements it is meant to qualify are held to.

11 Conclusion

An unresolved effect and an absent effect look identical in a table, and a project that publishes nulls as findings is precisely the project that cannot afford the confusion. FishAI published several before measuring what its instrument could resolve. Measured, the instrument's noise is 3.15–3.44 sets per duplicate pair, giving an 80%-power floor of 0.47 sets at $N = 400$ and 0.33 at $N = 800$; the normal-theory power model behind those figures was validated against a planted edge of known size in all twelve cells of both runs, on the strength of a byte-exact control that had first to be repaired, because the version that short-circuited at zero handicap proved only that the unwrapped policy equalled the shipped policy and never entered the wrapper at issue. Two parts did not replicate: the empirical floor at $N = 400$ is bounded only to 0.36–0.50, and interval coverage at $N = 400$ pools to 8.0% $[5.2, 11.7]$ against a nominal 5%, for reasons unknown and not assumed benign. The floor must always be that cell's own — one cell measures 3.645, a foreign-engine harness 4.57–4.95, and two mostly-agreeing arms resolve effects a generic floor would reject. Applied to the back catalogue, the floor turns two published nulls into non-measurements; the mechanisms the project ships clear their floors by a factor of three or more, with one exception the same rule catches — the weakest of §8.1's six triangle edges, plain against conceal-only, whose +0.1457 and +0.1742 sit at a floor of $\sim$0.15 and are nominal only.

The deliverable is not the number. It is the row the number belongs in: publish the floor beside the null.

References

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[2] FishAI project. *`scripts/probe-floor.mjs` — the detection-floor harness.* Source file; the four stages (`sd`, `calib`/`detect`, `fpr`, `published`), the planted-edge arm and its non-short-circuiting control, the

symnull and abnull true nulls, the MEASURED_SD constants recorded 2026-08-31 at 1,200 pairs per condition, the PUBLISHED claim list, and the NOMINAL ONLY -- underpowered verdict rule.

[3] FishAI project. *CROSSPLAY.md — FishAI against the fishlabs bots, in the fishlabs engine*. Repository document; §2 (the per-opponent cells at 150 games each, published without intervals), §3 (defusal replicated in a foreign engine, 300 duplicate pairs per opponent, half-widths 0.517–0.560).

[4] FishAI project. *BOUNDED.md — the bit-budget memory ladder*. Repository document; §5a (the v1.5-against-v1.0 ladder, 2,000 duplicate pairs per rung on bank v15v10-holdout-a, and the residual at an unbounded budget).

[5] FishAI project. *BOT_LAB.md — play-style spectrum: research, experimental design, and metrics*. Repository document; §5.1–5.2 (duplicate-deal pairing), §5.4 (fixed-N precision for published estimates), §5.5 (SPRT for tuning decisions only).

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[8] A. Hsieh. *FishAI v0.5: measuring play style without skill in a 54-card Literature variant*. Companion paper, FishAI project, 2026. The engine, the us54 dialect, the nine-style roster, and the tournament design every instrument in this paper is built on.

[9] A. Hsieh. *FishAI v1.5: a bit-budget memory ladder as an honest difficulty axis*. Companion paper, FishAI project, 2026. The generation the mostly-agreeing-arms discussion of Section 7.2 is set against. It does *not* supply that section's *v1.0*-against-*v0.5* cell, which appears in no committed document (Section 7.3); the traceable mostly-agreeing-arms measurement on this ladder is [4] §5a's v1.5-against-v1.0 rung.

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